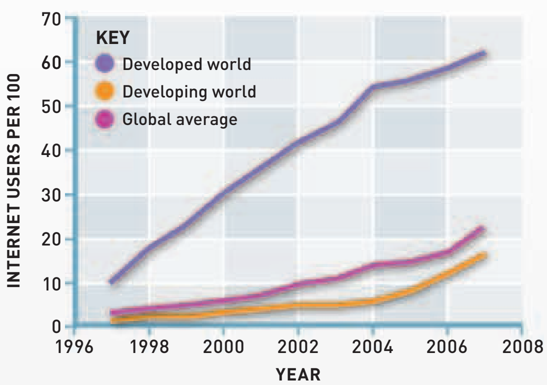
# THE STORY OF COMMUNICATION

ELECTRICITY TRIGGERS A REVOLUTION IN BROADCASTING AND PERSONAL COMMUNICATION


**Instant worldwide communication has become a defining characteristic of the modern world. Until 200 years ago, most long-distance messages could travel no faster than the horse or ship carrying them. It was the advent of electricity in the 19th century that transformed communications.**

In the 18th century, the French navy developed a system for transmitting orders between ships using semaphore flags. From the 1790s, semaphore was used on land, with lines of stations relaying coded messages using large signalling devices, each visible to the next station in the chain. From the 1830s, the development of electric telegraph replaced this medium. American Samuel Morse produced a robust and practical system, a simple on-off key generating a code that was transmitted along a wire. By the 1860s, telegraph wires spanned continents, and underwater cables enabled almost instant communication across oceans.

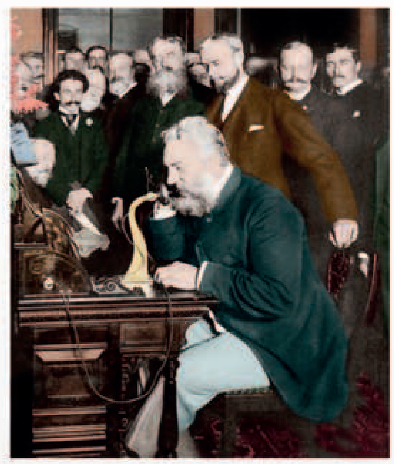
In the late 19th and early 20th centuries, the invention of the telephone enabled electronic transmission of speech. The discovery that radio waves could transmit sound opened the new possibility of broadcasting. From the 1920s, "wireless sets," providing entertainment and news, became a common feature of households. Television, however, did not become a mass medium until the 1950s.




**The rise of the Internet**
*The increase in Internet usage in developed countries was dramatic between 1997 and 2007, but access remained available only to a minority in the developing world.*

### THE INFORMATION AGE

Several lines of development revolutionized communications after World War II, creating the "Information Age." The advent of transistors made electronic goods smaller and cheaper, and advances in rocket technology allowed satellites to be placed in space, enabling global access to communication networks. The triumph of digital technology and microprocessors from the 1980s made computers almost universal. The potential flow of information worldwide was effectively limitless.



**Long-distance call**
*Alexander Graham Bell initiates the first telephone link between New York and Chicago in 1892. By then, New York was already linked to Boston and Philadelphia.*

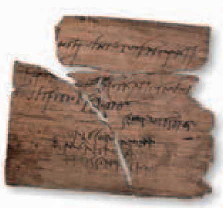
**Prehistory**
**Smoke signals**
Fire allows smoke signals to be sent over considerable distances. However, this method is limited to a simple set of prearranged messages.

Sumerian tablet

**3100–2500 BCE**
**Cuneiform writing**
Writing is a giant step forward in communication. Mesopotamian cuneiform script is inscribed on clay tablets.

**2900–2350 BCE**
**Carrier pigeons**
Pigeons are used to carry messages in ancient Egypt and Persia. They will continue to be used by armies in World War I and World War II.

**1st–2nd centuries**
**Letters by courier**
Letters are written on papyrus or on wood in the Roman Empire. The Vindolanda tablets from Roman Britain include a birthday invitation.



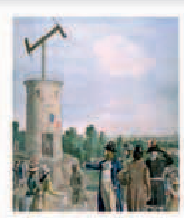
Vindolanda tablet

**17th century**
**Newspapers**
Newspapers, which disseminate information to a large public, develop in 17th-century Europe. The development of the printing press contributes to their growth.

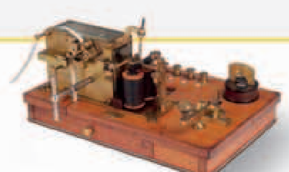

*The London Post*

**1784**
**Mail coaches**
Britain introduces four-horse coaches that are faster than passenger-carrying stagecoaches to carry post between major cities.

**1791–95**
**Visual telegraphy**
French inventor Claude Chappe pioneers a semaphore system that allows coded messages to be transmitted by chains of relay stations.



Chappe telegraph



Morse receiver

**1837**
**Electric telegraphy**
American inventor Samuel Morse develops the electric telegraph in the US. British Railways uses an electric telegraph.